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S2C2Seg: Semantic-Spatial Consistency and Category Optimization for Open-Vocabulary Segmentation

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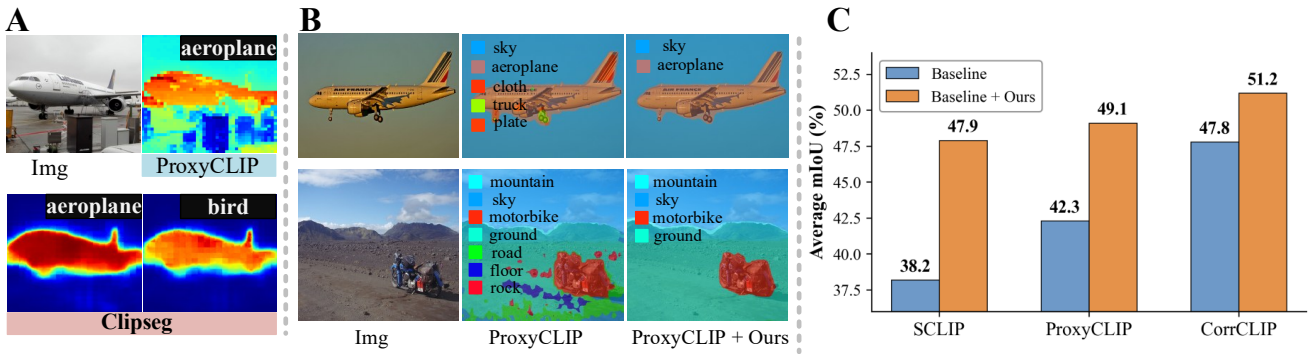


Figure 1. **Motivation, Framework, and Performance Comparison of Open-Vocabulary Segmentation.** **(A)** Existing methods suffer from spatial confusion and category ambiguity, causing inaccurate segmentation of similar or adjacent objects. **(B)** S2C2Seg selects high-confidence category subsets and applies consistent semantic guidance for accurate, category-aware segmentation. **(C)** Mean mIoU (%) across eight benchmarks. S2C2Seg improves SCLIP, ProxyCLIP, and CorrCLIP by +9.7, +6.8, and +3.4 points respectively.

Abstract

Open-vocabulary semantic segmentation extends pixel-level recognition to arbitrary text-described categories. Despite strong global semantic understanding, vision-language models such as CLIP exhibit limited spatial precision and semantic ambiguity across large vocabularies, constraining their effectiveness for dense prediction. We present S2C2Seg, a training-free framework that integrates with existing methods through Category Subset Selection (CSS) and Consistent Semantic Guidance (CSG). CSS employs multi-aspect scoring to filter candidate categories: CLIP-based global semantic similarity, spatial presence from dense prediction models, and cross-view consistency via alignment and conditional entropy. This joint exploitation of semantic, spatial, and consistency cues reduces category redundancy and semantic ambiguity. CSG adaptively fuses CLIP global features with local spatial predictions through category-specific confidence weighting, applying stronger regularization to high-similarity categories for correcting prediction biases while preserving spatial precision for low-confidence categories. Extensive experiments across eight benchmarks demonstrate broad applicability: when integrated with SCLIP, ProxyCLIP, and CorrCLIP, S2C2Seg achieves consistent improvements of 3.4 to 9.7 percentage points in mIoU, establishing a new state-of-the-art of 51.2% average mIoU.

1. Introduction

Vision-language models such as CLIP [31] demonstrate strong zero-shot recognition across arbitrary text-specified categories, yet extending it to open-vocabulary semantic segmentation [30] remains challenging. CLIP's contrastive pretraining optimizes global image-text alignment, yielding

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strong classification but limited spatial granularity for dense prediction [25, 42]. Recent studies [25, 42] identify two primary challenges: coarse spatial localization in attention maps originally designed for classification, and overlapping activations among semantically similar or co-occurring categories when scaled to large vocabularies, degrading segmentation quality.

Existing methods address these challenges through two primary directions. Spatial refinement approaches enhance localization through CLIP attention refinement [3, 13, 25, 42], feature denoising [1, 22, 35], or integration of complementary models such as DINO [5], diffusion models [34]. Despite improving spatial precision, these methods treat all candidate categories uniformly regardless of semantic relationships or prediction confidence, propagating ambiguous activations from global features to final predictions when large vocabularies contain similar categories such as airplane and aircraft or co-occurring objects such as road and vehicle. Disambiguation approaches [2, 18, 41] reduce semantic overlap through similarity-based category ranking or selection, constraining vocabulary size yet relying on global similarity metrics without explicitly modeling spatial prediction consistency. As these two directions target spatial and semantic challenges respectively, the interdependence between coarse localization and category overlap suggests that joint modeling may yield superior results.

We propose S2C2Seg, a training-free framework coordinating spatial localization and semantic disambiguation. Image-level models like CLIP [31] provide robust global semantics but lack spatial precision, while pixel-level models like CLIPSeg [26] enable fine-grained localization yet yield inconsistent cross-category predictions. S2C2Seg reconciles these complementary limitations via two coordinated components: Category Subset Selection (CSS) filters candidate categories via multi-aspect scoring, reducing redundancy and ambiguity; Consistent Semantic Guidance (CSG) fuses global semantic features with local spatial predictions through category-specific confidence weighting, with stronger regularization for semantically proximate categories while preserving spatial precision. Figure 1 illustrates improvements across eight benchmarks with representative baselines.

Our primary contributions are as follows:

- We propose a training-free framework that integrates global semantic models and local spatial predictors to jointly address vocabulary disambiguation and spatial refinement in open-vocabulary segmentation, achieving consistent improvements across diverse baseline methods without additional training.
- We introduce CSS, which filters candidate categories through multi-aspect scoring that integrates global similarity, spatial presence, and cross-view consistency, reducing category redundancy and semantic ambiguity.
- We develop CSG, which adaptively fuses global semantic features with local spatial predictions via category-specific confidence weighting, applying stronger regularization to semantically proximate categories while preserving spatial precision.

2. Related Work

2.1. Open-Vocabulary Segmentation

Open-vocabulary semantic segmentation extends closed-set segmentation to arbitrary text-described categories. **Training-based methods** enhance CLIP's segmentation capability through region-text pseudo-labeling [8, 15], image-level weak supervision [12, 14, 27], mask proposal generation [49, 50], and cross-modal distillation [24, 53], with recent advances further addressing decoding efficiency and cross-domain generalization [16, 23, 52, 56]. Despite strong in-domain performance, these methods typically require task-specific annotations. **Training-free methods** use pre-trained VLMs without fine-tuning for improved generalization to novel categories. Representative strategies include self-attention refinement [10, 22, 25, 42] that modifies CLIP's internal attention for better spatial localization, external prior integration from self-supervised or diffusion models [21, 36, 40] for enhanced spatial detail, and ensemble [3, 13] or global-local consistency strategies [43]. S2C2Seg addresses category space optimization through multi-dimensional filtering and enhances prediction quality via category-aware adaptive fusion of global and local features.

2.2. Category Selection Strategies

Category selection remains underexplored in OVSS. Existing methods rely on single-dimensional criteria: CaR [41] refines categories via CLIP similarity, potentially underweighting those with strong local presence but weak global alignment. FLOSS [2] uses text template entropy with predefined templates, while CDAM [18] measures spatial attention distribution through JS divergence for category relevance. DenseCLIP [32] and ViewCo [33] explore dense prediction and multi-view consistency; Mask-Adapter [24] enforces mask IoU consistency but requires training. In contrast, CSS integrates three complementary dimensions, namely global semantic alignment, local spatial activation, and prediction entropy, within a two-stage framework that balances recall and precision for dynamic category filtering.

2.3. Semantic Guidance for Dense Prediction

Multiple approaches exploit CLIP's semantic knowledge for dense prediction: category-level modulation [18, 23, 32, 38] employs CLIP features as fixed semantic priors, adaptive text guidance [44, 45] dynamically modulates embeddings conditioned on image context, and recent meth-

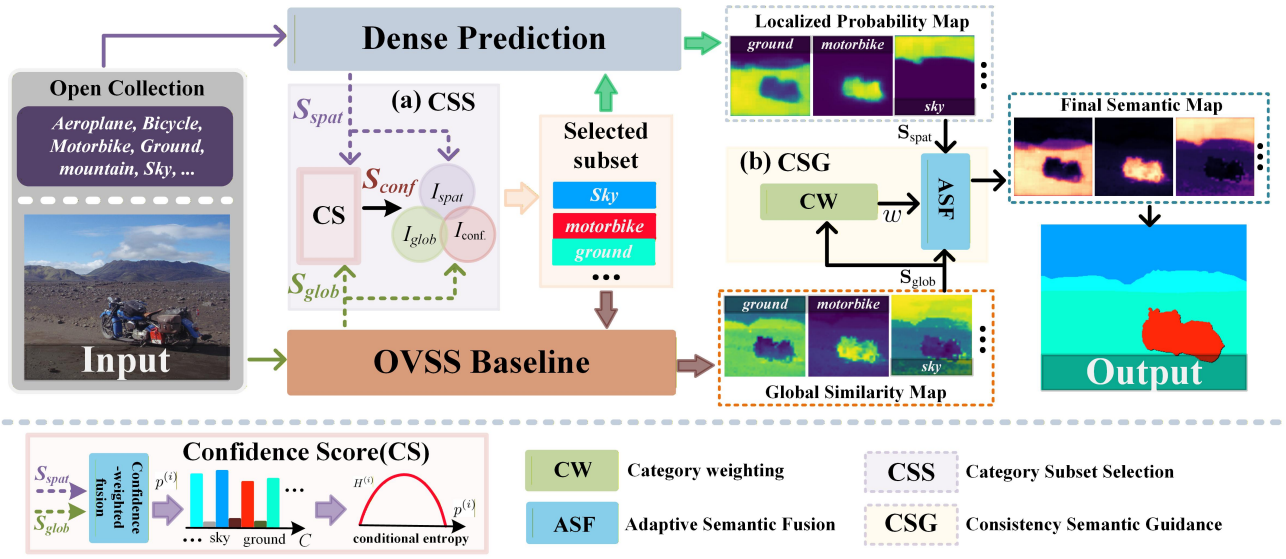


Figure 2. S2C2Seg framework architecture. Given an input image and vocabulary $\mathcal{T}$, **(a)** CSS filters $\mathcal{T}$ to a compact subset $\mathcal{T}'$ through multi-aspect scoring of global CLIP similarity, local CLIPSeg response, and cross-view confidence with adaptive thresholding. **(b)** CSG generates final segmentation via category-aware adaptive fusion of global and local features with confidence weighting.

ods further explore learnable fusion with multi-scale integration [29]. Our CSG mechanism introduces category-aware dynamic weighting that modulates local predictions via CLIP's per-category global confidence, enabling adaptive trust allocation across the vocabulary while preserving spatial precision under strong semantic priors.

3. Method

3.1. Overview

Given an image $\mathbf{I} \in \mathbb{R}^{H\times W\times 3}$ and C semantic categories $\mathcal{C} = \{c_1, \ldots, c_C\}$ with text descriptions $\mathcal{T} = \{t_1, \ldots, t_C\}$, open-vocabulary semantic segmentation produces a dense prediction $\mathbf{M} \in \{1, \ldots, C\}^{H\times W}$ assigning each pixel to one category. Standard OVSS methods independently evaluate all C categories for each pixel, leading to category redundancy where predictions scatter across visually similar categories, and global-local inconsistency where pixel-wise predictions lack coherent semantic constraints. We propose a two-stage framework: Category Subset Selection filters $\mathcal{C}$ to $\mathcal{C}' \subset \mathcal{C}$ with $K = |\mathcal{C}'|$, and Consistent Semantic Guidance refines predictions via adaptive fusion. Given baseline F_{base}, our segmentation is given by

$$\mathbf{M} = \text{CSG}(\mathbf{I}, \mathcal{C}', \mathbf{S}_{\text{spat}}), \tag{1}$$

where $\mathbf{S}_{\text{spat}}$ denotes pixel-level spatial predictions for the filtered subset.

3.2. Category Subset Selection

Given full category set $\mathcal{C}$, CSS selects $\mathcal{C}' \subset \mathcal{C}$ maximizing segmentation quality while minimizing redundancy through multi-aspect scoring and adaptive thresholding.

3.2.1. Multi-aspect Scoring

Three complementary functions assess each category c_i: global semantic alignment measures image-level category matching through vision-language embeddings, local spatial presence quantifies fine-grained evidence from dense prediction activations, and cross-view consistency validates reliability via global-local agreement following multi-view learning principles.

Global Semantic Alignment. We employ CLIP encoders [31] to generate text embeddings $\mathbf{T} = [\mathbf{t}_1, \ldots, \mathbf{t}_C]^\top \in \mathbb{R}^{C\times d}$ and global image feature $\mathbf{v}_{\text{glob}} \in \mathbb{R}^d$. We compute the global similarity score $s^{(i)}_{\text{glob}}$ for category c_i via L_2-normalized cosine similarity between $\mathbf{v}_{\text{glob}}$ and text embedding $\mathbf{t}_i \in \mathbb{R}^d$.

Local Spatial Presence. To complement global semantic alignment with fine-grained spatial evidence, we quantify the local presence of each category through dense spatial activations. A dense prediction model generates pixel-level activation maps $\mathbf{P}^{(i)} \in [0,1]^{H'\times W'}$ for each category c_i, where $H' \times W'$ denotes the activation map resolution. The spatial presence score $s^{(i)}_{\text{spat}}$ is obtained by averaging activations across all spatial locations, with higher values indicating stronger evidence for category c_i presence.

Cross-view Confidence. We quantify prediction confidence through cross-view alignment following multi-view learning principles. Given score vectors $\mathbf{s}_{\text{glob}}, \mathbf{s}_{\text{spat}} \in \mathbb{R}^C$, we apply L_1 normalization to obtain distributions $\bar{s}^{(i)}_{\text{glob}} = s^{(i)}_{\text{glob}} / \sum_j s^{(j)}_{\text{glob}}$ and $\bar{s}^{(i)}_{\text{spat}}$ similarly. The fusion weight α is

derived from their inner product:

$$\alpha = \sigma\left(\sum_{i=1}^{C} \bar{s}_{\text{glob}}^{(i)} \cdot \bar{s}_{\text{spat}}^{(i)} - 0.5\right), \tag{2}$$

where $\sigma(\cdot)$ denotes the sigmoid function. This formulation assigns higher weight when both views agree and lower weight when they conflict. The fused distribution is constructed through weighted averaging and renormalization as

$$p^{(i)} = \frac{\alpha \bar{s}_{\text{glob}}^{(i)} + (1-\alpha)\bar{s}_{\text{spat}}^{(i)}}{\sum_{j=1}^{C}\left[\alpha \bar{s}_{\text{glob}}^{(j)} + (1-\alpha)\bar{s}_{\text{spat}}^{(j)}\right]}. \tag{3}$$

Category-wise confidence is quantified via conditional entropy. After selecting category c_i, we compute the residual distribution $p_{\text{res}}^{(j|i)} = p^{(j)}/(1-p^{(i)})$ for $j \neq i$ and obtain normalized conditional entropy as

$$H^{(i)} = -\frac{1}{\log(C-1)} \sum_{j \neq i} p_{\text{res}}^{(j|i)} \log p_{\text{res}}^{(j|i)}. \tag{4}$$

The final confidence score combines selection probability with certainty:

$$s_{\text{conf}}^{(i)} = p^{(i)}(1 - H^{(i)}) \in [0,1]. \tag{5}$$

This captures both presence likelihood via $p^{(i)}$ and choice certainty via $1 - H^{(i)}$, favoring categories with high probability and low ambiguity.

3.2.2. Two-stage Filtering Strategy

To maximize recall while reducing redundancy, we employ a two-stage filtering strategy. Multi-aspect Aggregation constructs an initial set $\mathcal{C}_{\text{init}}$ with high recall, then Adaptive Size Control refines it to the final subset $\mathcal{C}'$.

Multi-aspect Aggregation. To ensure comprehensive coverage across scoring perspectives, we independently select top-ranked categories from each score vector with unified retention ratio $\tau \in (0,1]$ as

$$\mathcal{I}_k = \text{TopK}(\mathbf{s}_k, \lfloor \tau \cdot C \rfloor), \quad k \in \{\text{glob}, \text{spat}, \text{conf}\}, \tag{6}$$

The initial category subset is then obtained by taking the union of the three index sets as

$$\mathcal{C}_{\text{init}} = \{c_i : i \in \mathcal{I}_{\text{glob}} \cup \mathcal{I}_{\text{spat}} \cup \mathcal{I}_{\text{conf}}\}. \tag{7}$$

This union operation maximizes recall by retaining categories with strong evidence in any aspect, while subsequent Adaptive Size Control refines the subset through cross-aspect reranking.

Adaptive Size Control. To ensure $K_{\min} \leq |\mathcal{C}'| \leq K_{\max}$, we rerank categories in $\mathcal{C}_{\text{init}}$ using a unified scoring function. For each category $c_i \in \mathcal{C}_{\text{init}}$, we apply min-max normalization to each score type $k \in \{\text{glob}, \text{spat}, \text{conf}\}$ to obtain normalized scores $\hat{s}_k^{(i)} \in [0,1]$. The final ranking score is obtained by summing the three normalized components as

$$s_{\text{final}}^{(i)} = \hat{s}_{\text{glob}}^{(i)} + \hat{s}_{\text{spat}}^{(i)} + \hat{s}_{\text{conf}}^{(i)}, \quad c_i \in \mathcal{C}_{\text{init}}. \tag{8}$$

The output subset $\mathcal{C}'$ consists of top-K categories ranked by $s_{\text{final}}^{(i)}$, where K is bounded by $K_{\min} = 6$ and $K_{\max} = 20$. The lower bound ensures sufficient category diversity for simple scenes, while the upper bound prevents excessive redundancy in complex scenarios.

3.3. Consistent Semantic Guidance

Given filtered category subset $\mathcal{C}'$ with $K = |\mathcal{C}'|$ categories from CSS, CSG refines pixel-level predictions by adaptively fusing global semantic features with local spatial predictions under category-specific confidence weighting.

3.3.1. Dual-stream Feature Extraction

We extract two complementary visual representations that provide global context and local details respectively.

Global Semantic Features. The CLIP visual encoder extracts patch-level features $\mathbf{V} \in \mathbb{R}^{N \times d}$, where $N = H_p W_p$ denotes the number of spatial patches. Let $\mathbf{T}' = [\mathbf{t}'_1, \ldots, \mathbf{t}'_K]^\top \in \mathbb{R}^{K \times d}$ denote text embeddings for categories in $\mathcal{C}'$. We compute patch-text similarity via cosine similarity. Let $\bar{\mathbf{V}} \in \mathbb{R}^{N \times d}$ and $\bar{\mathbf{T}}' \in \mathbb{R}^{K \times d}$ denote row-wise L_2 normalized versions of $\mathbf{V}$ and $\mathbf{T}'$. The similarity matrix is

$$\mathbf{S}_{\text{glob}} = \bar{\mathbf{V}} \bar{\mathbf{T}}'^{\top} \in \mathbb{R}^{N \times K}. \tag{9}$$

We reshape $\mathbf{S}_{\text{glob}}$ and upsample it from patch resolution $H_p \times W_p$ to image resolution $H \times W$ via bilinear interpolation, yielding $\mathbf{S}_{\text{glob}} \in \mathbb{R}^{K \times H \times W}$.

Local Spatial Predictions. To obtain fine-grained spatial localization, we employ dense prediction model CLIPSeg generating pixel-level prediction logits. While excelling at capturing spatial details, these local predictions lack global semantic consistency, necessitating fusion with global features. Formally, the dense model processes image $\mathbf{I}$ and text descriptions $\mathcal{T}'$ for categories in $\mathcal{C}'$:

$$\mathbf{S}_{\text{spat}} = F_{\text{dense}}(\mathbf{I}, \mathcal{T}') \in \mathbb{R}^{K \times H \times W}, \tag{10}$$

where F_{dense} denotes any dense prediction model such as CLIPSeg.

3.3.2. Adaptive Semantic Fusion

To jointly combine global semantic consistency with local spatial precision, we fuse $\mathbf{S}_{\text{glob}}$ and $\mathbf{S}_{\text{spat}}$ using category-specific confidence weighting. For each category $c_i \in \mathcal{C}'$, we first compute the spatial average of global similarity scores as

$$g^{(i)} = \frac{1}{HW} \sum_{h=1}^{H} \sum_{w=1}^{W} \mathbf{S}_{\text{glob}}^{(i)}(h,w). \tag{11}$$

Table 1. Quantitative comparison on eight benchmarks. mIoU (%) averaged across all datasets. **With background**: VOC21, Ctx60, COCO; **Without background**: VOC20, City, Ctx59, ADE, Stuff.

Method	Venue	With background			Without background					Avg. ↑
		VOC21	Ctx60	COCO-Obj	VOC20	City	Ctx59	ADE	Stuff	
GroupViT[48]	CVPR'22	50.4	18.7	27.5	79.7	11.1	23.4	9.2	15.3	27.7
TCL[6]	CVPR'23	51.2	24.3	30.4	77.5	23.1	30.3	14.9	19.6	33.1
CLIP-Dinoiser[47]	ECCV'24	62.2	32.4	35.0	80.2	31.7	35.9	20.0	24.6	40.3
CoDe[46]	CVPR'24	57.5	30.5	32.3	-	28.9	-	17.7	23.9	-
CLIP[31]		20.8	9.3	8.9	49.1	6.7	11.2	3.2	5.7	14.4
ReCo[37]	NeurIPS'22	25.1	19.9	15.7	57.7	21.6	22.3	11.2	14.8	23.7
MaskCLIP[9]	ICML'23	38.8	23.6	20.6	74.9	12.6	26.4	9.8	16.4	27.9
CLIPSurgery[25]	PR'25	55.2	30.3	29.7	77.5	33.1	33.4	16.1	22.2	37.2
GEM[3]	CVPR'24	46.2	32.6	33.9	79.9	21.2	35.9	15.7	23.7	36.1
ClearCLIP[22]	ECCV'24	57.0	32.2	32.5	82.3	32.8	35.8	17.3	24.0	39.2
CaR[41]	CVPR'24	48.6	13.6	15.4	73.7	–	18.4	5.4	–	–
OVDiff[19]	ECCV'24	66.3	29.7	34.6	80.9	23.4	32.9	14.1	20.3	37.8
LAVG[17]	ECCV'24	62.1	31.6	34.2	82.5	26.2	34.7	15.8	23.2	38.8
CDAM[18]	ICLR'25	58.7	30.6	35.2	–	23.7	–	17.2	24.8	–
ResCLIP[51]	CVPR'25	61.1	33.5	35.0	86.0	35.9	36.8	18.0	24.7	41.4
SCCLIP[1]	TIP'25	64.6	36.8	37.7	84.3	41.0	40.1	20.1	26.6	43.9
Trident[36]	ICCV'25	67.1	38.6	41.1	84.5	42.9	42.2	21.9	28.3	45.8
LPOSS+[39]	CVPR'25	62.4	35.4	34.3	79.3	37.9	38.6	22.3	26.5	42.1
CASS[20]	CVPR'25	65.8	36.7	37.8	87.8	39.4	40.2	20.4	26.7	44.4
SCLIP[42]	ECCV'24	59.1	30.4	30.5	80.4	32.2	34.2	16.1	22.4	38.2
+Ours	–	**70.7**	**41.8**	**40.5**	**88.9**	**41.3**	**47.7**	**22.4**	**30.0**	**47.9** ↑9.7
ProxyCLIP[21]	ECCV'24	61.3	35.3	37.5	80.3	38.1	39.1	20.2	26.5	42.3
+Ours	–	**72.8**	**42.7**	**41.5**	**89.2**	**43.6**	**48.7**	**23.1**	**31.0**	**49.1** ↑6.8
CorrCLIP[54]	ICCV'25	71.9	41.1	40.7	87.8	42.0	46.3	22.2	30.5	47.8
+Ours	–	**75.7**	**44.5**	**44.8**	**91.1**	**47.0**	**50.8**	**23.7**	**32.3**	**51.2** ↑3.4

This aggregation quantifies semantic strength for category c_i and modulates local predictions $\mathbf{S}^{(i)}_{\text{spat}}$ to integrate global context with spatial detail. The scores $\mathbf{g} = [g^{(1)}, \ldots, g^{(K)}]^\top$ are transformed into confidence weights $\mathbf{w} = [w^{(1)}, \ldots, w^{(K)}]^\top$ via temperature-scaled softmax. The fused logits for category c_i are then computed as

$$\mathbf{S}^{(i)}_{\text{fused}} = \mathbf{S}^{(i)}_{\text{glob}} + \lambda \cdot w^{(i)} \cdot \mathbf{S}^{(i)}_{\text{spat}}, \quad i = 1, \ldots, K, \tag{12}$$

where λ balances the contribution of global semantic features and locally weighted spatial predictions and $w^{(i)}$ provides adaptive category-specific weighting: categories with stronger global semantic evidence receive higher weights for local feature integration, while weak categories are down-weighted to reduce noise.

Finally, to generate predictions over the original C categories, we construct $\mathbf{S}_{\text{final}} \in \mathbb{R}^{C \times H \times W}$ by mapping the K filtered categories back to their positions in the full label space:

$$\mathbf{S}^{(j)}_{\text{final}}(h, w) = \begin{cases} \mathbf{S}^{(i)}_{\text{fused}}(h, w) & \text{if } c_j \in \mathcal{C}', \\ -\infty & \text{otherwise}, \end{cases} \tag{13}$$

where i denotes the index of category c_j within $\mathcal{C}'$. The large negative value effectively excludes filtered-out categories from consideration. The final segmentation map $\mathbf{M} \in \{1, \ldots, C\}^{H \times W}$ is obtained via per-pixel argmax: $\mathbf{M}(h, w) = \arg\max_j \mathbf{S}^{(j)}_{\text{final}}(h, w)$.

4. Experiments

4.1. Experimental Setup

Datasets and Metrics. We evaluate on eight benchmarks: PASCAL VOC [11] (20/21 classes), PASCAL Context [28] (59/60 classes), COCO-Object [4] (80 classes), COCO-Stuff [4] (171 classes), ADE20K [55] (150 classes), and Cityscapes [7] (19 classes). We report mean Intersection-over-Union (mIoU, %) on validation sets.

Implementation Details. We implement our framework in PyTorch on NVIDIA RTX 4090 GPUs. We adopt ViT-B/16 CLIP [31] as the visual-language backbone and CLIPSeg [26] for dense prediction. Following standard practice [31], we use 80 prompt templates per category for text embedding. For CSS, we set retention ratio $\tau = 0.3$ and subset size bounds $K_{\min} = 6$, $K_{\max} = 20$. For CSG,

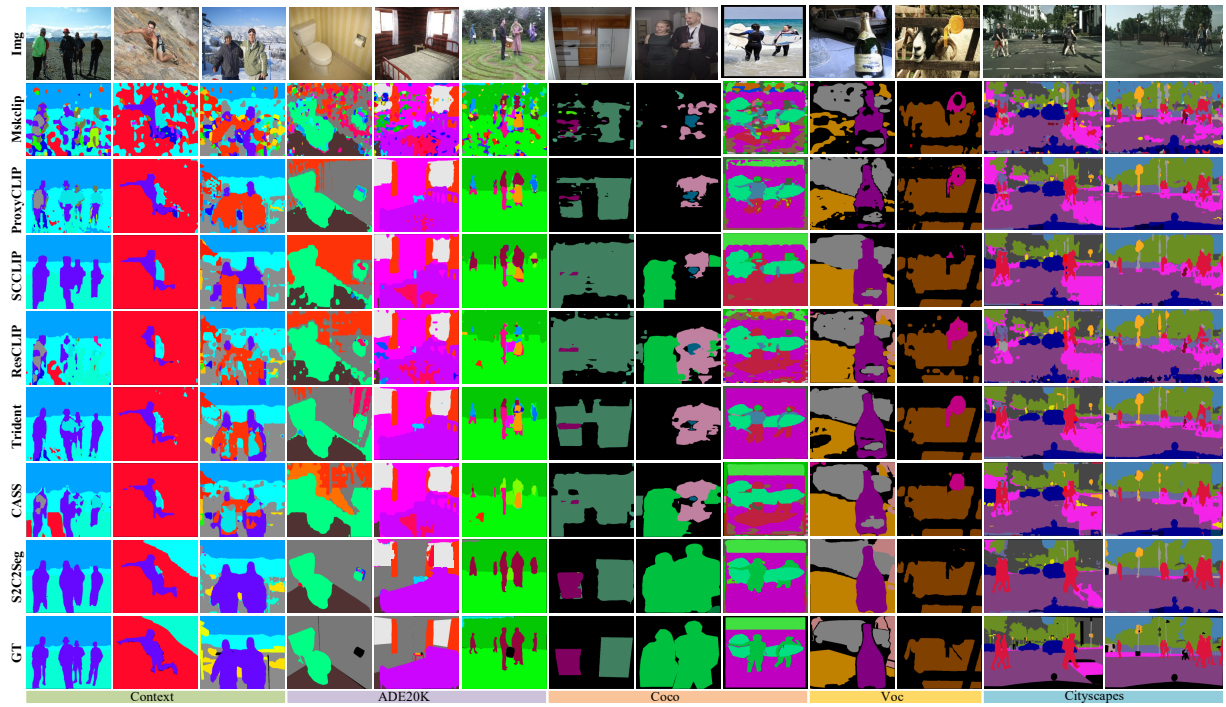


Figure 3. Qualitative comparison on PASCAL Context, ADE20K, COCO-Object, PASCAL VOC, and Cityscapes. Our S2C2Seg produces more accurate segmentations with improved category discrimination and spatial consistency compared to state-of-the-art methods.

Table 2. Ablation study on framework components. CSS refines category subsets; CSG performs category-aware fusion. Evaluated on eight benchmarks with two baselines (ProxyCLIP and CLIPSeg). **w/o CSG**: uniform weighting replaces category-aware.

Method	With background			Without background				
	VOC21	Ctx60	COCO	VOC20	City	Ctx59	ADE	Stuff
Baselines								
ProxyCLIP	61.3	35.3	37.5	80.3	38.1	39.1	20.2	26.5
CLIPSeg	63.5	36.2	37.1	76.1	39.2	41.8	21.3	25.3
With CSS (category refinement)								
ProxyCLIP	64.3	37.2	38.9	84.8	39.7	40.7	20.9	28.1
CLIPSeg	66.3	37.9	39.0	80.7	40.4	42.9	21.7	27.7
Proposed framework variants								
Ours (w/o CSG)	70.4	41.0	39.3	87.6	41.8	44.1	20.4	28.8
Ours (w/o CSS)	68.0	39.5	39.1	87.5	40.2	42.5	21.7	28.3
Ours (Full)	**72.8**	**42.7**	**41.5**	**89.2**	**43.6**	**48.7**	**23.1**	**31.0**

we set fusion weight $\lambda = 0.6$. Images are resized such that the shorter side is 336 pixels for PASCAL VOC and Context datasets, and 448 pixels for ADE20K, Cityscapes, and COCO-Stuff. To demonstrate generalizability, we integrate S2C2Seg with three representative baselines employing distinct feature extraction strategies: SCLIP [42] refines attention mechanisms, ProxyCLIP [21] incorporates self-supervised models, and CorrCLIP [54] represents current state-of-the-art performance.

Table 3. Ablation of CSS scoring dimensions across eight benchmarks. **Oracle (GT)**: uses only the ground-truth categories present in each image with our CSG module.

Config.	With background			Without background				
	VOC21	Ctx60	COCO	VOC20	City	Ctx59	ADE	Stuff
w/o Sel.	68.0	39.5	39.1	87.5	40.2	42.5	21.7	28.3
S_{glob} only	68.8	40.1	39.8	88.0	40.7	43.2	22.0	29.3
$S_{\text{glob}}+S_{\text{spat}}$	70.1	41.2	40.7	88.6	42.1	46.3	22.8	30.1
$S_{\text{glob}}+S_{\text{conf}}$	70.5	40.9	40.5	88.3	42.4	46.8	22.4	30.3
Full (Ours)	**72.8**	**42.7**	**41.5**	**89.2**	**43.6**	**48.7**	**23.1**	**31.0**
Oracle (GT)	80.5	57.2	52.2	95.6	57.3	65.8	38.3	48.6

4.2. Main Results

4.2.1. Quantitative Analysis

Table 1 reports mIoU across eight benchmarks. S2C2Seg+CorrCLIP achieves 51.2% mIoU, surpassing Trident at 45.8% and CASS at 44.4% by 5.4 and 6.8 points. Integration with SCLIP, ProxyCLIP, and CorrCLIP yields gains of +9.7, +6.8, and +3.4 points, where improvement inversely correlates with baseline sophistication. On Context59 with 59 categories, CSS reduces the category space to achieve 50.8% mIoU. On VOC21, where background accounts for 21 categories, CSG exploits CLIP's global discriminative capacity to correct local biases from background interference, yielding +11.5 points.

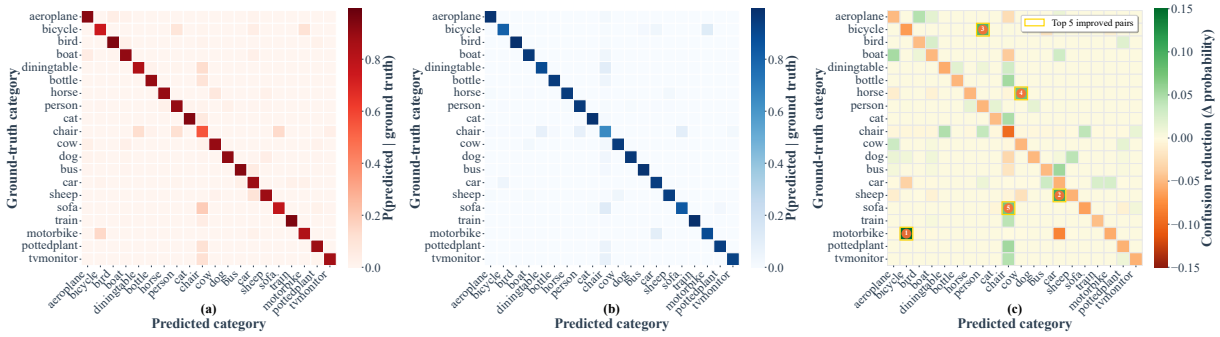


Figure 4. Category confusion analysis on PASCAL VOC20 validation set. (a) Confusion matrix for ProxyCLIP baseline showing frequent confusion between semantically similar categories (e.g., bicycle/motorbike, car/bus). (b) Confusion matrix for ProxyCLIP + S2C2Seg showing reduced off-diagonal confusion. (c) Confusion reduction heatmap (green indicates improvement); yellow annotations highlight the top-5 category pairs with greatest confusion reduction.

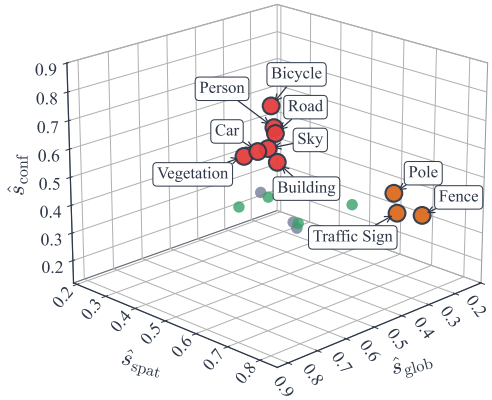


Figure 5. 3D visualization of CSS category scoring on PASCAL VOC20. Each candidate is plotted by global semantic score S_{glob}, local spatial response S_{spat}, and prediction entropy S_{conf}. Selected categories are marked in red and orange; filtered ones in gray.

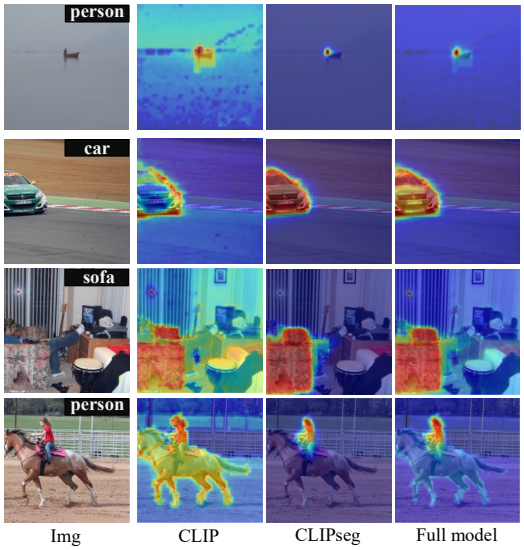


Figure 6. Visualization of complementary representations from CLIP global features and CLIPSeg local predictions.

4.2.2. Qualitative Analysis

Figure 3 presents qualitative comparisons across five benchmarks. On ADE20K indoor scenes in columns 4–6, our method produces spatially coherent segmentations where baselines exhibit fragmentation. Column 1 shows correct sky, ground, and person boundary segmentation; column 4 demonstrates precise toilet-room structure delineation. On COCO in columns 7–8 and VOC in column 11, S2C2Seg distinguishes table and cabinet from background while preserving person-background boundaries, confirming the complementary effects of CSS category refinement and CSG adaptive fusion.

4.3. Ablation Studies

4.3.1. Component Analysis

Table 2 evaluates framework components with ProxyCLIP and CLIPSeg baselines. CSS provides +3.0 mIoU on VOC21 and +4.5 on VOC20 through category refinement. **w/o CSG** achieves +9.1 on VOC21 via category refinement, while **w/o CSS** using all categories with adaptive fusion contributes +6.7. The full framework combining both modules achieves +11.5 on VOC21 and +8.9 on VOC20, where the VOC21 gain exceeds the sum of individual contributions (+3.0 and +6.7), demonstrating complementary benefits.

4.3.2. Multi-Dimensional Category Selection Analysis

Component-wise Ablation. Table 3 evaluates CSS scoring dimensions. Without category selection, performance averages 45.9% mIoU across eight benchmarks. Adding spatial score S_{spat} to global score S_{glob} improves performance to 47.7% mIoU, while adding confidence score S_{conf} yields 47.8%. Full CSS combining all three dimensions achieves

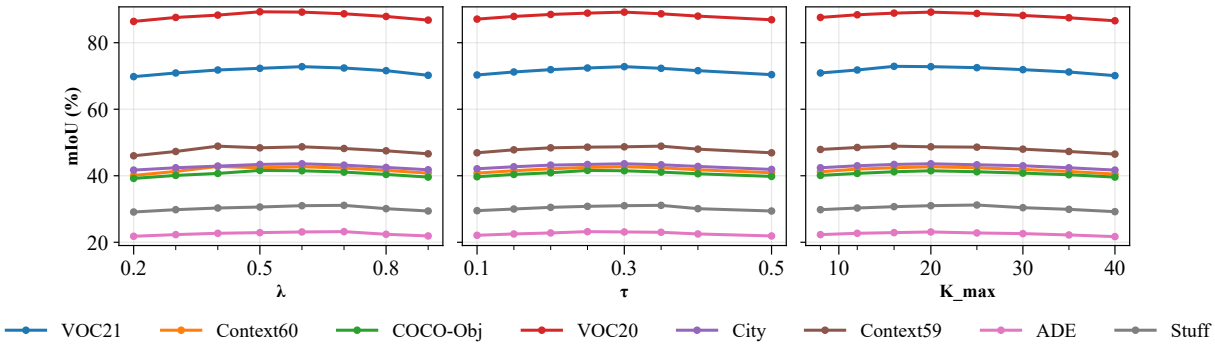


Figure 7. Hyperparameter sensitivity analysis. **(a)** Fusion weight λ in CSG module. **(b)** Upper bound $K_{\max}$ on selected category subset size. **(c)** Retention ratio τ in CSS union-based selection. Results demonstrate robust performance across parameter ranges.

Table 4. Comparison of fusion strategies in the CSG module (mIoU, %). All methods combine CLIP global semantic features $\mathbf{S}_{\text{glob}}$ with CLIPSeg local spatial features $\mathbf{S}_{\text{spat}}$ without training.

Method	With background			Without background				
	VOC21	Ctx60	COCO	VOC20	City	Ctx59	ADE	Stuff
Add.	70.4	41.0	39.3	87.6	41.8	44.1	20.4	28.8
Mult.	67.9	38.5	37.8	85.4	39.1	42.8	19.8	27.6
W.A.	69.6	39.8	39.7	86.8	42.1	45.7	21.1	28.5
MMN	68.1	39.0	38.4	86.2	41.5	44.9	21.3	28.1
Ours	**72.8**	**42.7**	**41.5**	**89.2**	**43.6**	**48.7**	**23.1**	**31.0**

49.1% mIoU, exceeding either pair and confirming complementary effects. Oracle with ground truth categories reaches 61.9%, with gap analysis in the supplementary document.

Confusion Matrix Analysis. Figure 4 examines confusion matrices on VOC20. Our method shows stronger diagonals and less off-diagonal confusion. CSS reduces confusion between related categories (bicycle/motorbike, car/bus). CSG reduces confusion between spatially adjacent categories (person/chair, person/table), maintaining spatial consistency via global guidance.

Scoring Space Visualization. Figure 5 visualizes the category distribution in CSS scoring space on VOC20. Gray points represent filtered categories with insufficient confidence or spatial evidence; blue points denote single-dimension-dominant categories; red and orange points indicate selected categories with balanced multi-dimensional scores. Selected categories concentrate in high-score regions where ground truth predominantly resides, confirming multi-dimensional scoring distinguishes reliable predictions from false positives.

4.3.3. Analysis of CSG Fusion Strategies

We compare five fusion strategies in Table 4: additive(Add.), multiplicative(Mult.), weighted averaging(W.A.), min-max normalization(MMN), and category-aware (ours). Category-aware fusion achieves 49.1% average mIoU, outperforming additive by +2.4, with notable gains on Context59 (+4.6), ADE (+2.7), and VOC21 (+2.4). Figure 6 visualizes the complementary nature of global and local representations. CLIP exhibits limited spatial precision (rows 1-2, person and boat regions overlap), while CLIPSeg provides finer localization (column 3). CSG modulates local evidence via global confidence, maintaining consistency when cues diverge (column 4).

4.4. Hyperparameter Sensitivity

We analyze three key hyperparameters on PASCAL VOC validation set using ProxyCLIP baseline (Figure 7). The fusion weight λ balances global semantic guidance with local spatial predictions. Performance remains stable at 48.0–49.5% mIoU across $\lambda \in \{0.4, 0.8\}$, and we adopt $\lambda = 0.6$. The retention ratio τ determines initial category set size before adaptive filtering, exhibiting robust performance across $\tau \in \{0.2, 0.5\}$ with $\tau = 0.3$ balancing recall and precision. The upper bound $K_{\max}$ constrains final subset size, showing consistent results for $K_{\max} \in \{15, 25\}$. We set $K_{\max} = 20$ based on typical object distributions.

5. Conclusion

We present S2C2Seg, a training-free framework for open-vocabulary semantic segmentation addressing category redundancy and global–local semantic inconsistency. CSS filters candidate categories via multi-aspect scoring to reduce redundancy and ambiguity while maintaining recall. CSG adaptively fuses CLIP's global semantic features with CLIPSeg's local spatial predictions via category-specific confidence weighting. Experiments across eight benchmarks show that integrating S2C2Seg with baselines such as SCLIP, ProxyCLIP, and CorrCLIP consistently improves performance by +3.4 to +9.7 pp mIoU, achieving state-of-the-art 51.2% mIoU.

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